

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457105.6684 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.047
Transit depth (Rp/Rs)^2	2.7 +/- 1.55 [%]
Orbital Inclination (inc)	84.67 +/- 2.84
Airmass coefficient 1 (a1)	0.159 +/- 0.099
Airmass coefficient 2 (a2)	1.49 +/- 1.45
Scatter in the residuals of the lightcurve fit is	63.41 %
Best Comparison Star	#1 - [428, 398]
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.082 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.047 vs 0.1368 (deviation 0.41σ)
Duration fit vs published	0.082 d vs 0.0758 d (deviation 0.24σ)
Mid-time O–C	3.2 min
Depth SNR	2.5
Residual scatter	63.41 %

Light curve

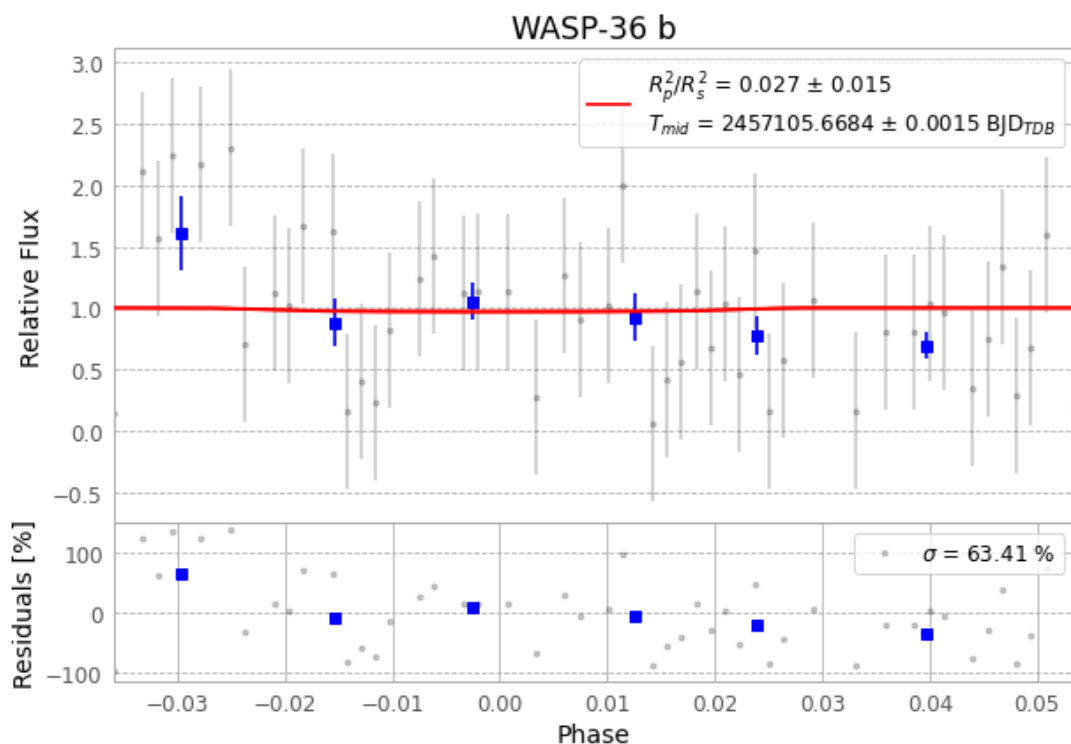


Figure 1 — FinalLightCurve_WASP-36 b_2015-03-23.png (SHA-256 850468c8c1d8e4e2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 363], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 07497cff7ca2014c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), WASP-36150324023612.FITS → WASP-36150324055412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-150324.log (922b8f7cb02b...), FinalLightCurve_WASP-36 b_2015-03-23.png (850468c8c1d8...), FinalLightCurve_WASP-36 b_2015-03-23.csv (6ebc84c52b71...)

Compute resources

Wall time 117.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:15Z.